

UBUNTU SERVER 26.04 LTS

PROFESSIONAL INSTALLATION MANUAL

A Complete Guide for System Administrators

Prepared by: Joya, Mazel Catherine P.

ITEP 414 - System Administration and Maintenance

BSIT 4C | Laguna State Polytechnic University

Instructor: John Randolph M. Penaredondo, MIT

Scenario: ABC Startup Solutions

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1. Introduction

This manual is written for system administrators who need to deploy an Ubuntu Server in a virtualized environment. It documents the exact procedures used to deploy the first server of ABC Startup Solutions, a startup company planning to host file sharing, remote administration, database hosting, web hosting, and internal services.

The manual is organized so that an administrator with no prior Ubuntu experience can follow it from start to finish. Every command and configuration used in this deployment is included, and each major step is accompanied by a screenshot.

2. Prerequisites and System Requirements

Requirement	Specification
Virtualization software	Oracle VirtualBox 7.x (or VMware Workstation)
Ubuntu Server ISO	Ubuntu Server 26.04 LTS (latest stable)
RAM	4 GB minimum
CPU	2 virtual processors
Storage	40 GB virtual disk (VDI)
Network	NAT (or Bridged)
Host system	Windows 11 with at least 8 GB RAM and VT-x enabled

3. Ubuntu Server Installation Guide

3.1 Creating the Virtual Machine

Open Oracle VirtualBox and click **New**. Enter the following settings:

Field	Value
Name	Ubuntu-Server-Week03
Type / Version	Linux / Ubuntu (64-bit)
Memory	4096 MB
Hard disk	Create a virtual hard disk now, 40 GB, VDI
Optical drive	Ubuntu Server 26.04 LTS ISO
Network	NAT

3.2 Installing Ubuntu Server

Start the virtual machine. The Subiquity installer will load. Follow these steps:

1. **Language:** select English.
2. **Keyboard layout:** select English (US).
3. **Network configuration:** accept DHCP and record the assigned IP address.
4. **Hostname:** enter `server01`.
5. **User account:** create a non-root administrative user with a strong password.
6. **Disk partitioning:** select Guided - Use an Entire Disk, enable LVM.
7. **SSH server:** enable Install OpenSSH Server.
8. **Additional packages:** do not install any unless instructed.
9. **Complete installation:** restart the VM and remove the ISO.

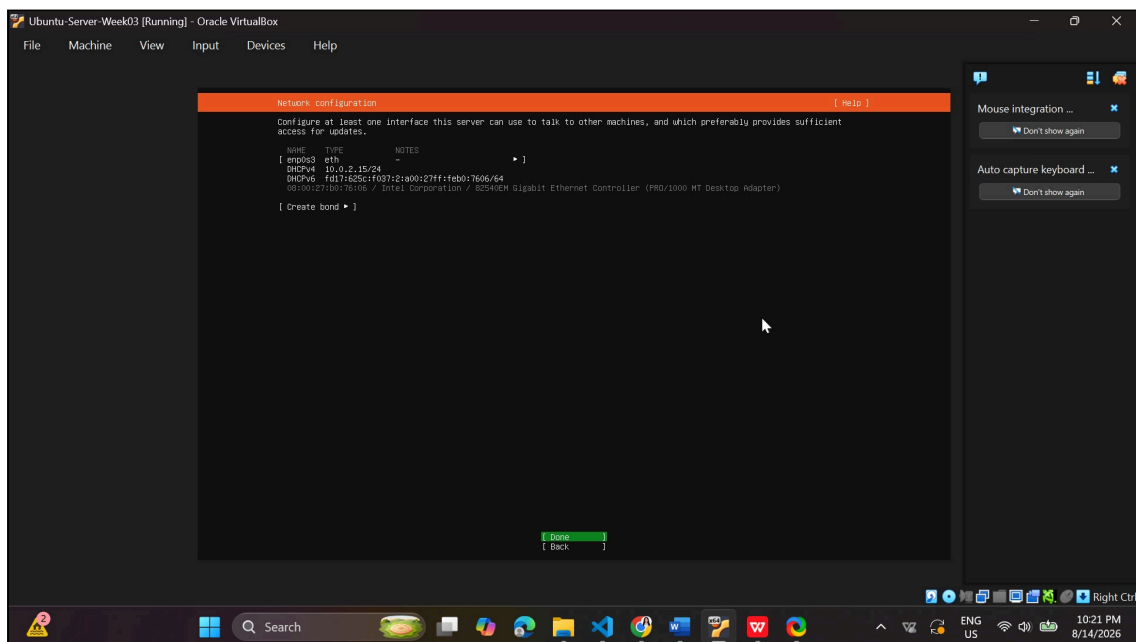


Figure 1. Network configuration with DHCP-assigned address 10.0.2.15/24

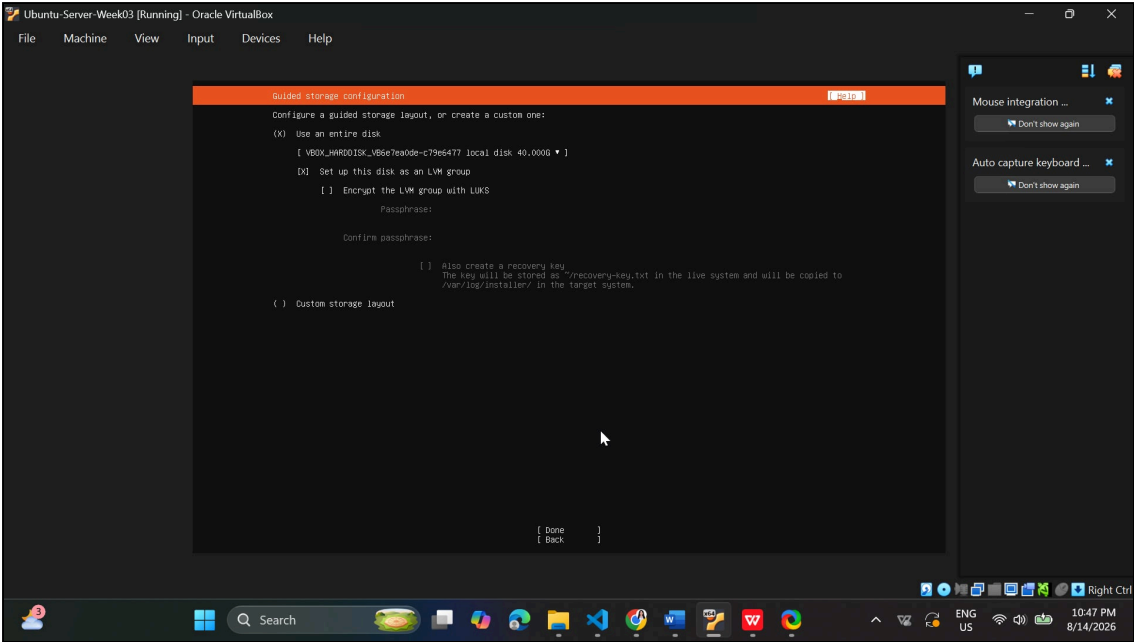


Figure 2. Guided storage configuration using the entire disk with LVM

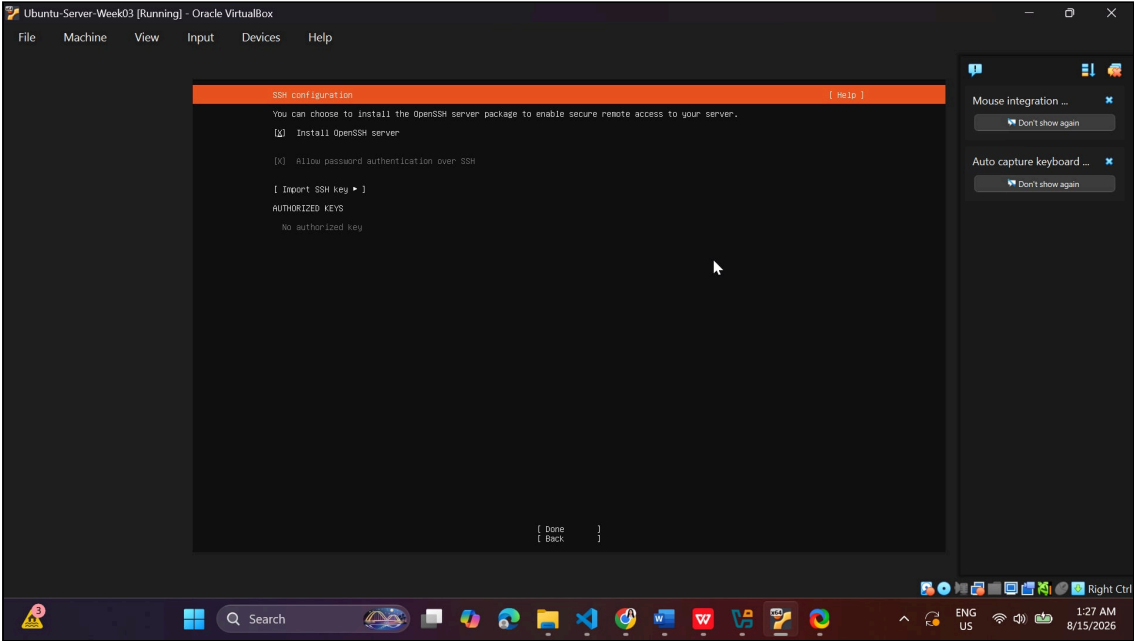


Figure 3. OpenSSH server enabled

4. Initial Server Configuration

4.1 First Login

Log in with the account created during installation. The message of the day will display the Ubuntu version and basic system information.

```

[ 79.203033] cloud-init[978]: |+ =
[ 79.208775] cloud-init[978]: | +. + 0
[ 79.210217] cloud-init[978]: |**% = o S
[ 79.217308] cloud-init[978]: | =0.8 . = .
[ 79.219252] cloud-init[978]: |+.B o 0 0
[ 79.226230] cloud-init[978]: |.00. * +
[ 79.227889] cloud-init[978]: |. 0+0
[ 79.229405] cloud-init[978]: |-----[SHA256]-----+
[[[A^[[A^[[A^[[A^[[[A^[[A^[[A^[[A^[[A^[[[A^[[A^[[[A^[[[A^[[[
server01 login: cathy
Password:
Login incorrect

server01 login: cathy
Password:
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-29-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

System information as of Fri Aug 14 06:19:00 PM UTC 2026

System load:            0.25
Usage of /:              37.6% of 18.53GB
Memory usage:           7%
Swap usage:              0%
Processes:              114
Users logged in:         0
IPv4 address for enp0s3: 10.0.2.15
IPv6 address for enp0s3: fd17:625c:f037:2:a00:27ff:feb0:7606

Expanded Security Maintenance for Applications is not enabled.

48 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

cathu@server01:~$

```

Figure 4. First login to the server

4.2 Checking the Hostname

```
cathy@server01:~$ hostname
server01
```

```
cathy@server01:~$ hostname
server01
cathy@server01:~$ _
```

Figure 5. Hostname verification showing server01

4.3 Checking Network Settings

```
cathy@server01:~$ ip addr
```

Confirm that the `enp0s3` interface has the expected IP address (10.0.2.15/24 in this deployment).

```
cathy@server01:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:b0:76:06 brd ff:ff:ff:ff:ff:ff
    altname enx080027b07606
    inet 10.0.2.15/24 metric 100 brd 10.0.2.255 scope global dynamic enp0s3
        valid_lft 85849sec preferred_lft 85849sec
    inet6 fd17:625c:f037:2:a00:27ff:feb0:7606/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 86035sec preferred_lft 14035sec
    inet6 fe80:a00:27ff:feb0:7606/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
cathy@server01:~$ _
```

Figure 6. IP address verification

4.4 Updating the Server

```
cathy@server01:~$ sudo apt update
cathy@server01:~$ sudo apt upgrade -y
```

```
cathy@server01:~$ sudo apt update
Hit:1 http://ph.archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://ph.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:3 http://ph.archive.ubuntu.com/ubuntu resolute-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu resolute-security InRelease
53 packages can be upgraded. Run 'apt list --upgradable' to see them.
cathy@server01:~$ _
```

Figure 7. apt update refreshing package lists

```
cathy@server01:~$ sudo apt update
Hit:1 http://ph.archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://ph.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:3 http://ph.archive.ubuntu.com/ubuntu resolute-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu resolute-security InRelease
53 packages can be upgraded. Run 'apt list --upgradable' to see them.
cathy@server01:~$ sudo apt upgrade -y
The following packages were automatically installed and are no longer required:
  linux-headers-7.0.0-14      linux-image-unsigned-7.0.0-14-generic  linux-modules-7.0.0-14-generic  linux-tools-7.0.0-14-generic
  linux-headers-7.0.0-14-generic  linux-main-modules-zfs-7.0.0-14-generic  linux-tools-7.0.0-14      pollinate
Use 'sudo apt autoremove' to remove them.

Upgrading:
  apparmor                libapparmor1                linux-firmware-marvell-wireless  packagekit
  apport                  libfwupd3                   linux-firmware-mediatek          python3-apport
  apport-core-dump-handler libgcrpt20                  linux-firmware-mellanox-spectrum python3-distupgrade
  base-files              libpackagekit-glib2-18     linux-firmware-misc              python3-problem-report
  bind9-dnssutils         libsgutils2-1.48           linux-firmware-netronome         sg3-utils
  bind9-host              linux-firmware-amd-graphics linux-firmware-nvidia-graphics   sg3-utils-udev
  bind9-lbns              linux-firmware-amd-misc    linux-firmware-qlogic            ubuntu-kernel-accessories
  cloud-init              linux-firmware-broadcom-wireless linux-firmware-qualcomm-graphics  ubuntu-minimal
  cloud-init-base         linux-firmware-intel-graphics linux-firmware-qualcomm-misc      ubuntu-release-upgrader-core
  fwupd                  linux-firmware-intel-misc  linux-firmware-qualcomm-wireless  ubuntu-server
  gir1.2-packagekitglib-1.0 linux-firmware-intel-wireless linux-firmware-realtek             ubuntu-server-minimal
  iproute2               linux-firmware-marvell-prestera motd-news-config                  ubuntu-standard

Not upgrading yet due to phasing:
  grub-pc  grub-pc-bin  grub2-common  python3-software-properties  software-properties-common

Summary:
  Upgrading: 48, Installing: 0, Removing: 0, Not Upgrading: 5
  Download size: 584 kB / 713 MB
  Space needed: 1,198 kB / 10.6 GB available
```

Figure 8. apt upgrade installing 48 updated packages

Tip: Always update a freshly installed server before putting it into service. Updates close security vulnerabilities and package the fixes that matter.

5. Server Verification Procedures

Use these procedures to confirm the server is healthy before going live.

#	Check	Command	Expected Result
1	Hostname	hostname	server01
2	IP address	ip addr	10.0.2.15/24 on enp0s3
3	Internet connectivity	ping -c 4 google.com	0% packet loss
4	System updates	sudo apt update && sudo apt upgrade -y	All packages up to date
5	SSH service	systemctl status ssh	active (running)

```

cathy@server01:~$ ping -c 4 google.com
PING google.com (142.250.207.46) 56(84) bytes of data:
64 bytes from ncmnlh-ai-in-f14.1e100.net (142.250.207.46): icmp_seq=1 ttl=255 time=15.9 ms
64 bytes from ncmnlh-ai-in-f14.1e100.net (142.250.207.46): icmp_seq=2 ttl=255 time=9.60 ms
64 bytes from ncmnlh-ai-in-f14.1e100.net (142.250.207.46): icmp_seq=3 ttl=255 time=9.61 ms
64 bytes from ncmnlh-ai-in-f14.1e100.net (142.250.207.46): icmp_seq=4 ttl=255 time=12.5 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3007ms
rtt min/avg/max/mdev = 9.597/11.889/15.880/2.583 ms
cathy@server01:~$

```

Figure 9. Ping test with 0% packet loss

```

cathy@server01:~$ systemctl status ssh
• ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
   Active: active (running) since Fri 2026-08-14 18:04:57 UTC; 1h 19min ago
     Invocation: 12f8e803c34a40d8878d2cf67061a74a
   TriggeredBy: • ssh.socket
     Docs: man:sshd(8)
           man:sshd_config(5)
    Main PID: 1294 (sshd)
      Tasks: 1 (limit: 1688)
     Memory: 1.6M (peak: 2.3M)
        CPU: 102ms
     CGroup: /system.slice/ssh.service
             └─1294 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Aug 14 18:04:56 server01 systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Aug 14 18:04:57 server01 sshd[1294]: Server listening on 0.0.0.0 port 22.
Aug 14 18:04:57 server01 systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
Aug 14 18:04:57 server01 sshd[1294]: Server listening on :: port 22.
cathy@server01:~$

```

Figure 10. SSH service active (running)

6. Troubleshooting Guide

Problem 1: "Login incorrect" on first login

The password was mistyped. Linux credentials are case-sensitive and are not echoed on screen.

Solution: type the username and password carefully and try again.

Problem 2: No internet connectivity

- Verify the VM network adapter is set to NAT or Bridged in VirtualBox settings.
- Run `ip addr` to confirm an address was assigned. If only 127.0.0.1 shows, the interface is down.
- Run `sudo netplan apply` or restart the VM to renew DHCP.

Problem 3: SSH service not running

```
cathy@server01:~$ sudo systemctl enable --now ssh
```

This enables and starts the SSH service. Check the status again with `systemctl status ssh`.

Problem 4: VM will not boot from ISO

- Check the storage settings and confirm the ISO is attached to the optical drive.
- Confirm the boot order places the optical drive before the hard disk.
- Check that the ISO file is not corrupted by verifying its SHA-256 checksum.

Problem 5: Ubuntu installer says no disk found

- Make sure the virtual disk was created (not just the ISO attached).
- In the installer, refresh the storage devices list.
- For SATA/NVMe issues, try switching the storage controller type in VM settings.

7. Best Practices

1. **Always use a non-root user.** Never run day-to-day commands as root. Use `sudo` only when needed.
2. **Use strong passwords.** A server exposed to the internet will be attacked within minutes; weak credentials are an invitation.
3. **Keep the system updated.** Run `apt update` and `apt upgrade` regularly, and enable automatic security updates if possible.
4. **Use SSH keys.** For production, disable password authentication and use key-based SSH authentication only.
5. **Document everything.** Record hostnames, IP addresses, and configuration changes so the next administrator can pick up where you left off.
6. **Back up configuration files.** Before editing files in `/etc`, copy them (e.g., `sudo cp file file.bak`).
7. **Test in a VM first.** A virtual machine is the safest place to break things and learn.
8. **Monitor the server.** Check service statuses and logs (`journalctl -u ssh`) periodically.

Important: In this project, no additional packages were installed and password authentication was left as configured by the installer, following the module instructions. For a production deployment, hardening steps such as firewall configuration (UFW) and SSH key-only authentication are strongly recommended.

8. References

1. Canonical. Ubuntu Server documentation. <https://ubuntu.com/server/docs>
2. Oracle. VirtualBox User Manual. <https://www.virtualbox.org/manual>
3. OpenSSH. OpenSSH Manual Pages. <https://www.openssh.com/manual.html>
4. CompTIA. Server+ (SK0-005) Exam Objectives. <https://www.comptia.org/certifications/server>